

13 — Structural Comparison: Previous vs. Current Project Version

Overview of Platform Evolution

This document provides a comparative analysis of the transformation from the **Previous Project Architecture** (generic maritime vessel tracker dashboard) to the **Current Project Architecture (Predictive Cargo Analytics & Decision Support System)**.

Comprehensive Structural Comparison Table

Functional Area	Previous Project Version	Current Enhanced Project Version	Why Changed	Major Benefit / Business Value
Primary System Focus	Generic vessel tracking & ship ETA counting	■ Predictive Cargo Analytics & Volume Forecasting	Vessels are inputs; ports care about cargo throughput & terminal capacity	Direct decision support for terminal operators & hinterland industries
Data Foundation	Mock vessel AIS location coordinates	Actual NMPA monthly cargo records (1,354 rows, 2021–2026)	Synthetic vessel dots lack real-world grounding	Mathematically grounded in real Indian port operations
ML Algorithms	Simple linear trendline extrapolation	Holt-Winters Exponential Smoothing + Multi-Feature Ridge Regression	Linear models fail to capture monsoon seasonality	Accurate 3, 6, and 12-month forecasts with 95% dynamic prediction bounds
Model Validation	No backtesting or accuracy measurement	Chronological out-of-sample backtesting (WAPE, MAPE, MAE, RMSE vs Naive)	Mentors & operators demand proof of ML model reliability	Verified performance metrics (WAPE 17.1%, accuracy 82.8%)
Explainability	Black-box numerical outputs	Quantified driver weight decomposition (MRPL Crude, UPCL Coal, Monsoon)	Operators must know WHY a forecast is rising/falling	Justified allocation of crane crews, tugboats, and berth depths
Scenario Testing	Static UI with fixed controls	Interactive What-If Cargo Scenario Simulator	Real ports face sudden delays and traffic spikes	Instant stress-testing of throughput & capacity risk levels
Trade Intelligence	Generic trade route listing	CODE_PH_0 with YoY growth tracking	Connects global trade lanes directly to port commodity demand	Identifies high-margin trade lanes & revenue growth corridors
Anomaly Engine	Basic threshold alerts	CODE_PH_0 with severity progression scale	Surges and declines require tiered response protocols	Visual classification (CODE_PH_0)

AI Capabilities	None / Generic static text	LangGraph 3-Tier Cognitive Dispatcher & AI Maritime Copilot	Natural language query dispatch across specialized agents	Instant answer synthesis for operational staff
UI Aesthetics	Default Streamlit layout with basic tables	Premium Warm Amber / Cream design system with Plotly theme	Professional visual presentation for executive demos	High-impact dashboard experience
Backend Architecture	Monolithic single script	Modular FastAPI microservice framework with 12 API routers	Monoliths are hard to maintain and scale	Clean separation of routing, analytics, and data pipeline layers
Data Ingestion	Fixed dataset file reading	Centralized Pandas pipeline + Multipart CSV Upload Endpoint (CODE_PH_0)	Ports need ability to ingest new monthly reports	Flexible ingestion for dynamic port data updates
NMPA Infrastructure	Generic terminal labels	Specific berth assignments, draft depths (meters), & hinterland consumers	Ports operate around physical draft and berth constraints	Realistic operational mapping (Oil Jetties, JSW Terminal, UPCL)